HARD · ADVANCED MATH

Advanced Math

11

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GRID-IN ADVANCED MATH

$$2x^2 - 8x - 7 = 0$$

One solution to the given equation can be written as $\frac{8 - \sqrt{k}}{4}$, where k is a constant. What is the value of k?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2 ADVANCED MATH

$$fx = 59 - 2x$$

The function f is defined by the given equation. For which of the following values of k does $fk = 3k$?

A. $\frac{59}{5}$

B. $\frac{59}{2}$

C. $\frac{177}{5}$

D. 59

Q3 GRID-IN ADVANCED MATH

$$xkx - 56 = -16$$

In the given equation, k is an integer constant. If the equation has no real solution, what is the least possible value of k?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4 ADVANCED MATH

The function f is defined by $f x = ax^2 + bx + c$, where a, b, and c are constants. The graph of $y = f x$ in the xy-plane passes through the points $7, 0$ and $-3, 0$. If a is an integer greater than 1, which of the following could be the value of $a + b$?

A. -6

B. -3

C. 4

D. 5

Q5

GRID-IN ADVANCED MATH

If $3x^2 - 18x - 15 = 0$, what is the value of $x^2 - 6x$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

ADVANCED MATH

Let the function p be defined as $p(x)=\frac{(x-c)^2+160}{2c}$, where c is a constant. If $p(c)=10$, what is the value of $p(12)$?

A. 10.00

B. 10.25

C. 10.75

D. 11.00

Q7

GRID-IN ADVANCED MATH

$$y = -1.5$$

$$y = x^2 + 8x + a$$

In the given system of equations, a is a positive constant. The system has exactly one distinct real solution. What is the value of a?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

ADVANCED MATH

A park ranger hung squirrel houses each in the shape of a right rectangular prism for fox squirrels. Each house has a height of 11 inches. The length of each house's base is x inches, which is 1 inch more than the width of the house's base. Which function V gives the volume of each house, in cubic inches, in terms of the length of the house's base?

A. $Vx = 11xx$ - 1

B. $Vx = 11xx + 1$

C. $Vx = xx + 11x$ - 1

D. $Vx = xx + 11x + 1$

Q9

ADVANCED MATH

If $u-3=\frac{6}{t-2}$, what is t in terms of u ?

A. $t=\frac{1}{u}$

B. $t=\frac{2u+9}{u}$

C. $t=\frac{1}{u-3}$

D. $t=\frac{2u}{u-3}$

Q10

ADVANCED MATH

Two variables, x and y, are related such that for each increase of 1 in the value of x, the value of y increases by a factor of 4. When $x = 0$, $y = 200$. Which equation represents this relationship?

A. $y = 4x^{200}$

B. $y = 4200^x$

C. $y = 200x^4$

D. $y = 2004^x$